BACKGROUND

• Plant diseases cause up to 40% of global food production loss annually, driving demand for automated leaf image identification

• While deep learning models reach 98% accuracy on clean controlled datasets (PlantVillage), performance drops significantly on real-world field datasets (PlantDoc).

• Standalone architectures each have their own limitation: CNNs capture local textures but miss global context, while Vision Transformers capture global structure well but miss localised details.

PROJECT OBJECTIVE ANS SCOPE

OBJECTIVE

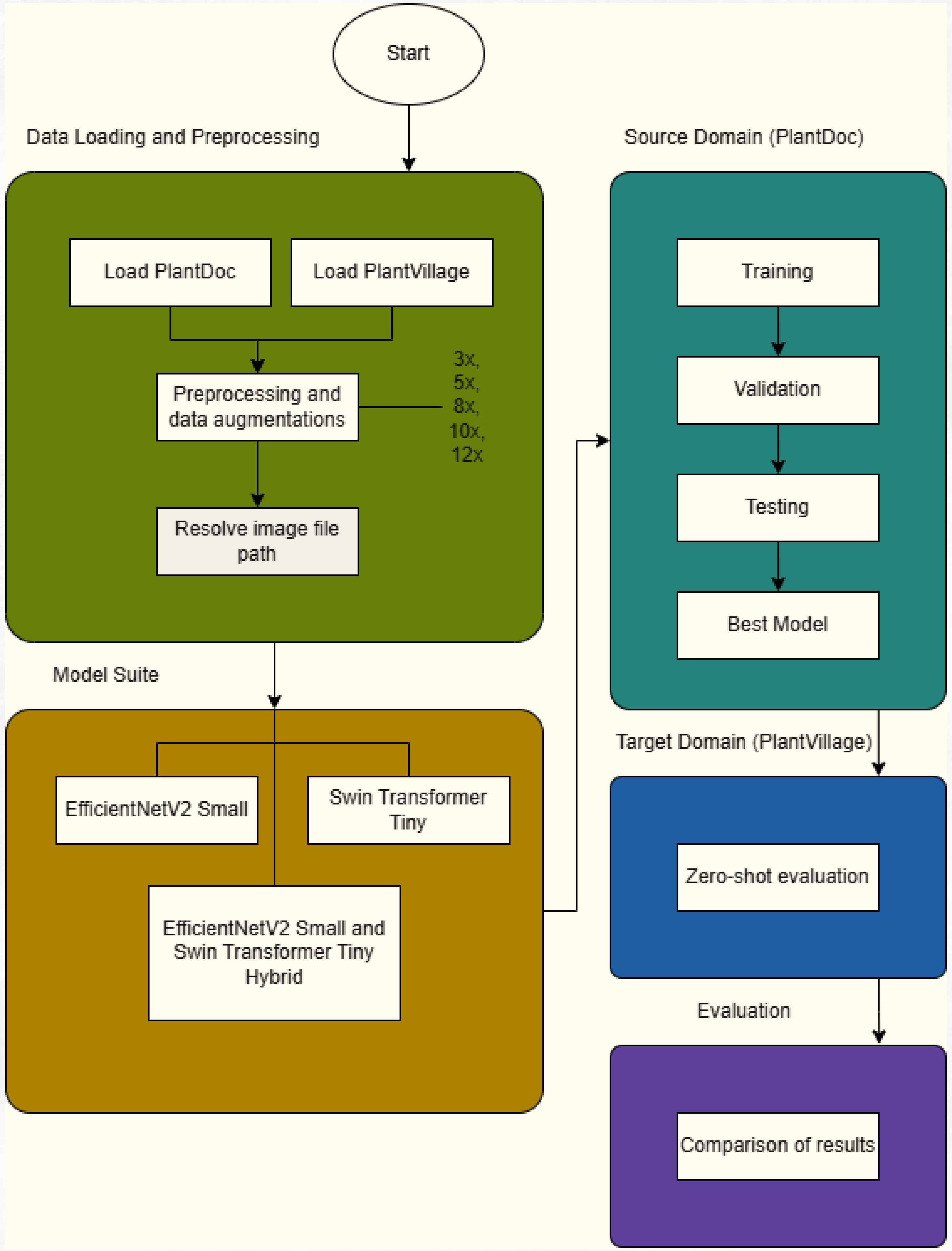
- Develop a custom hybrid architecture (EfficientNetV2 + Swin Transformer)
- Train on PlantDoc and evaluate zero-shot on PlantVillage to study cross-domain generalisation
- Evaluate the impact of increasing augmentation factors across three architectures

SCOPE

- Compare three models (EfficientNetV2, Swin Transformer, Hybrid CNN–Transformer) under the same hyperparameters
- Apply CLAHE preprocessing and data augmentation to address image quality
- Augmentation scaling (x1, x3, x5, x8, x10, x12) with zero-shot cross-dataset evaluation

METHODOLOGY: SYSTEM DESIGN OVERVIEW

SYSTEM DESIGN OVERVIEW



DATA PREPARATION AND PREPROCESSING

Training Dataset Augmentations (PlantDoc)

CLAHE

Random resizing

Random horizontal and vertical flipping

Random rotation

Colour jitters

Gaussian blurring

Random erasing

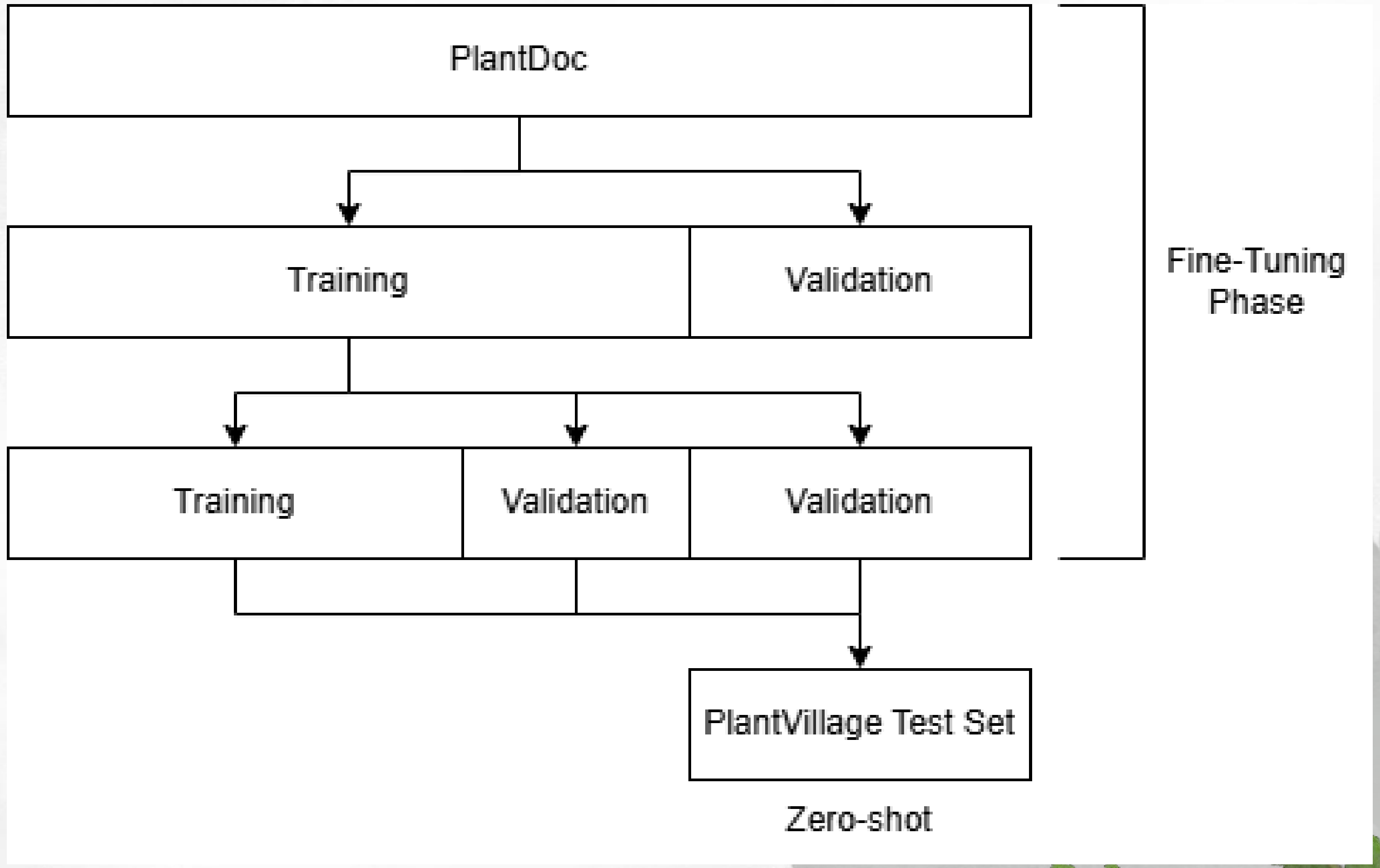
Normalisation

Validation and Testing Dataset Augmentations (PlantDoc and PlantVillage)

CLAHE

Resizing

Normalisation



Examples PlantDoc Training Set

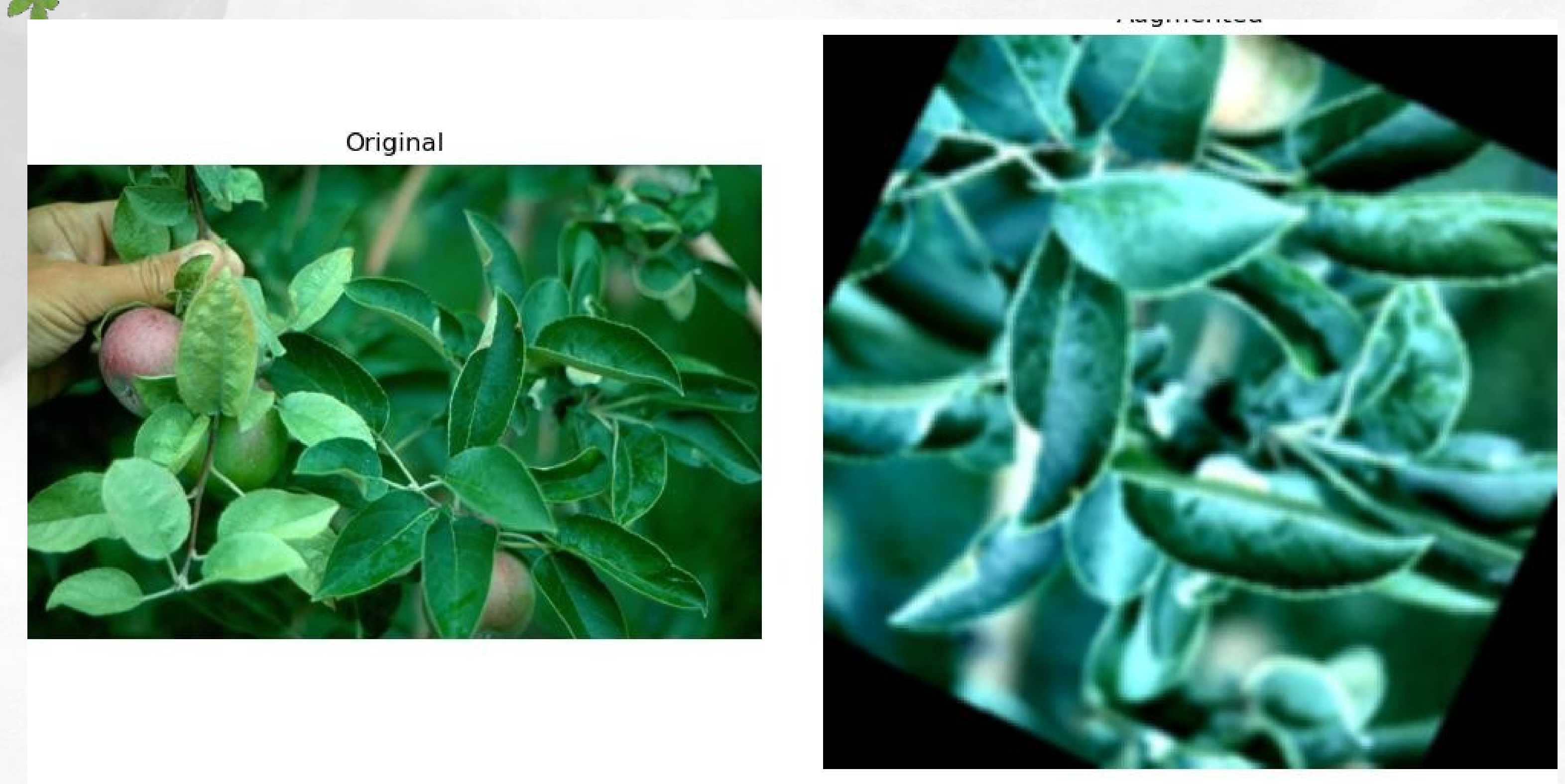


Figure 1

Figure 2

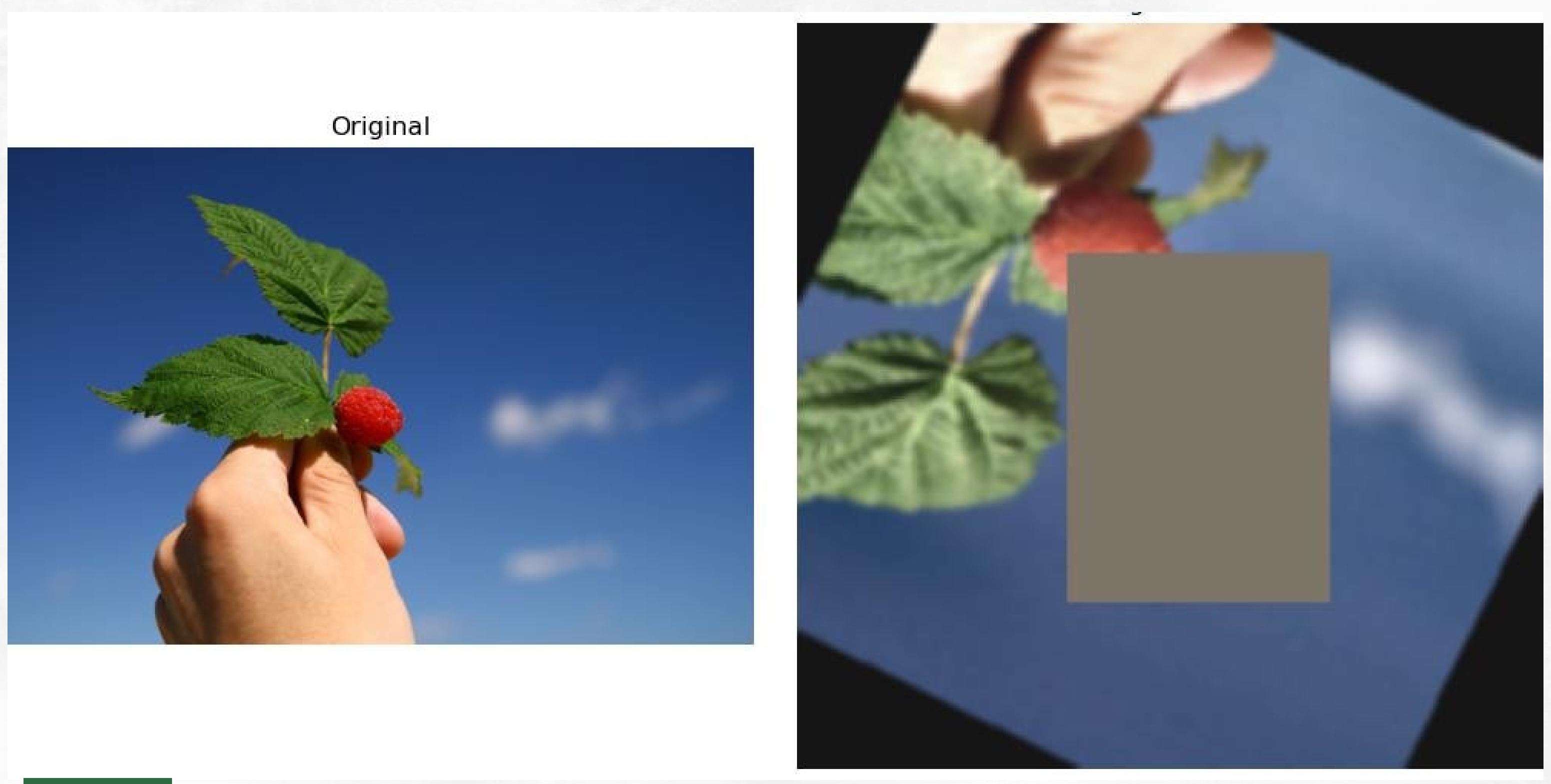


Figure 3

Examples PlantDoc Testing Set

Figure 1

Figure 2

Figure 3

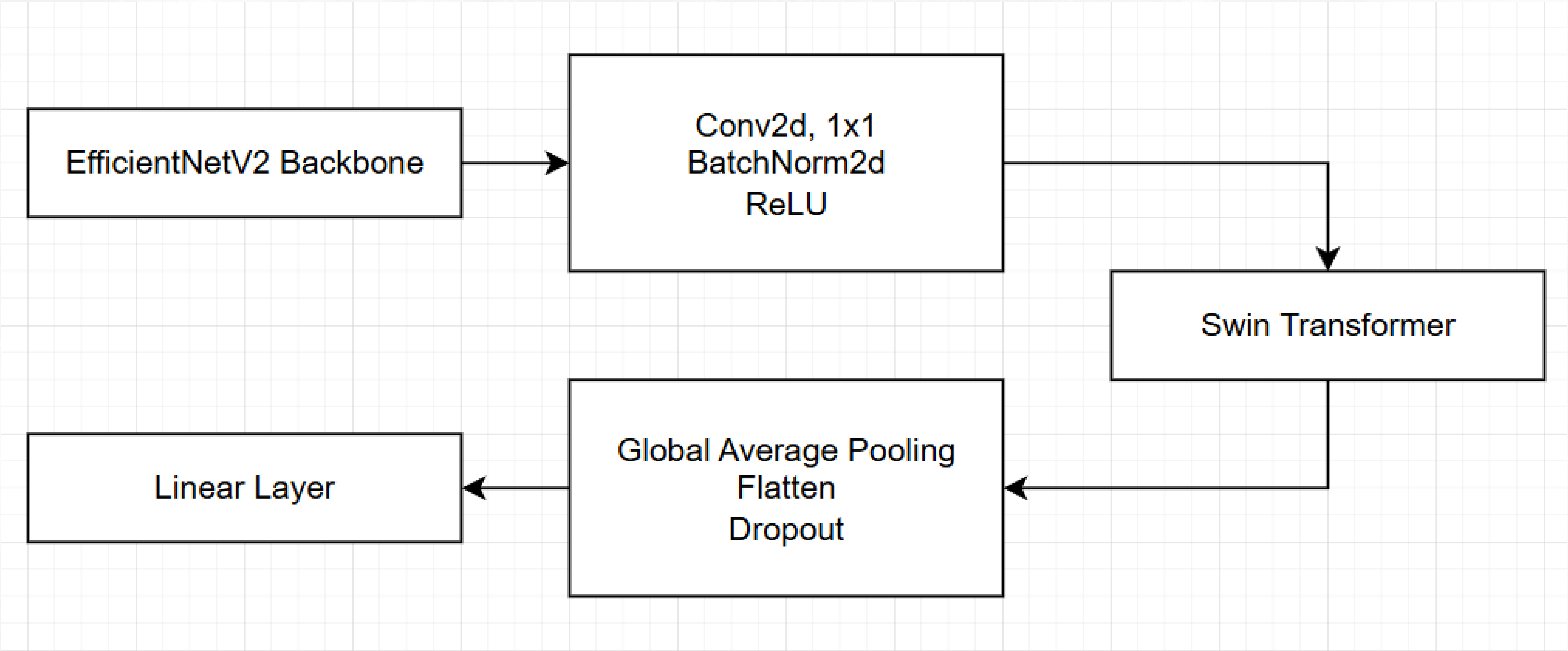
EfficientNetV2 and Swin

Insert the architecture

Talk about it

Split this if
you want

HYBRID ARCHITECTURE



Local inductive bias of EfficientNetV2

Global attention capability of Swin Transformer

TRAINING

Pipeline Configurations

Optimiser: AdamW

Scheduler: Linear warmup with cosine decay

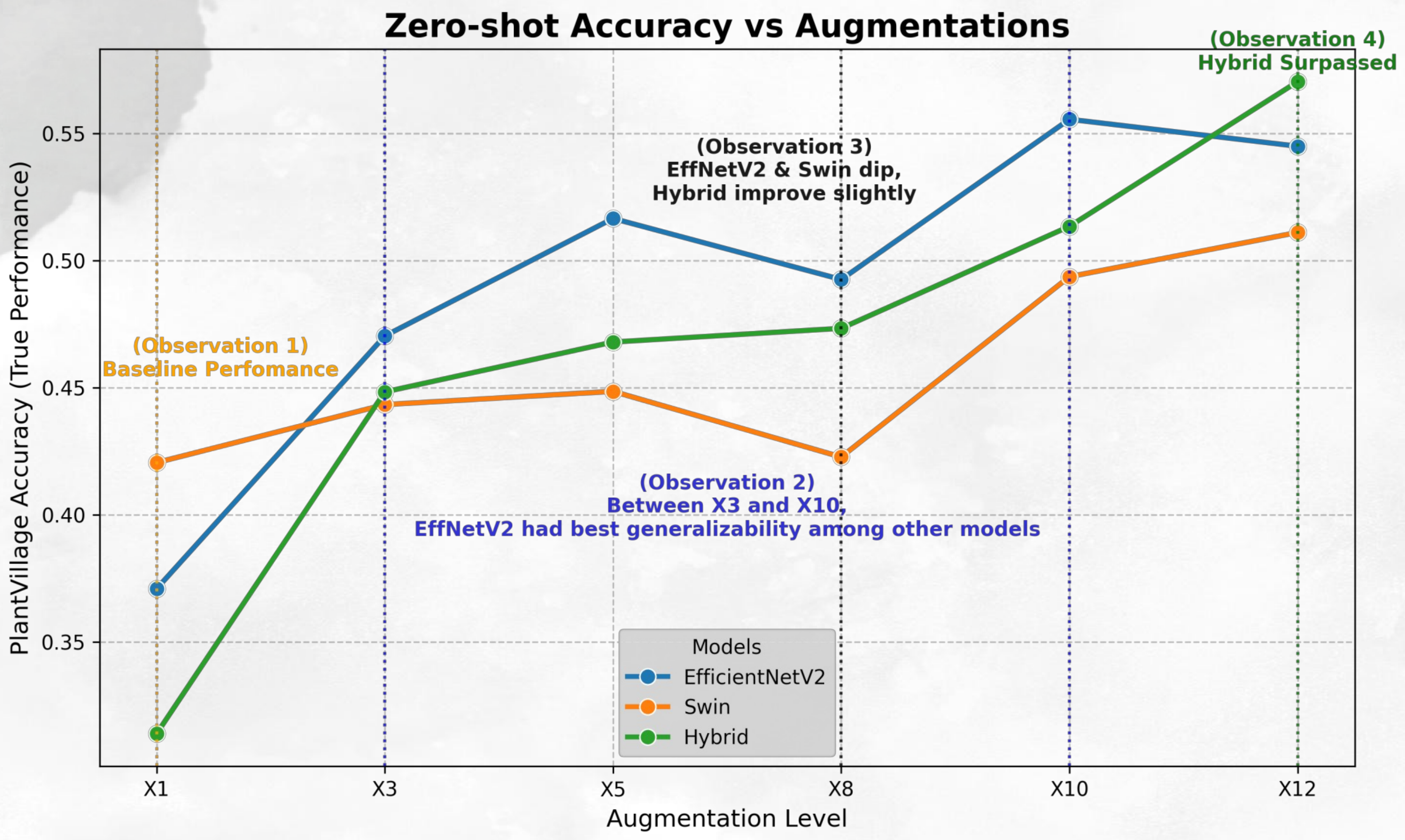
Learning rate: Differential learning rate

Early stopping: 10 epochs

Mixed Precision

EXPERIMENTAL RESULTS

Zero-Shot Results



Observation 1

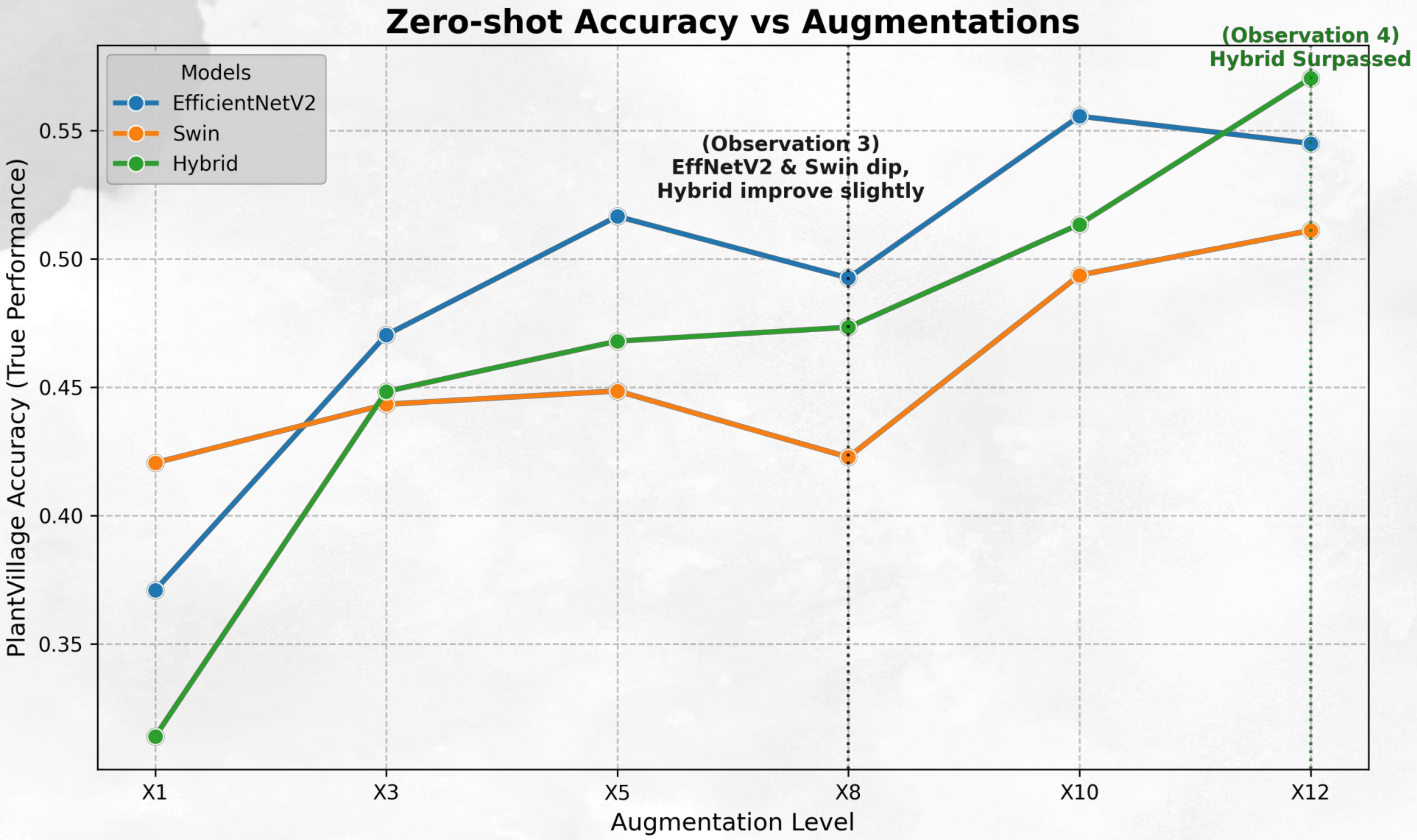
- Swin Transformer with x1 augmentation tops zero shot (0.42),
- **Advantage of global context** to isolate pathology from noises

Observation 2

- EfficientNetV2 dominate between x3 to x10 and peaking at x10 (0.56),
- Decline at x12 (0.54) as aggressive distortions disrupt the **local spatial context**

EXPERIMENTAL RESULTS

Zero-Shot Results (Cont'd)



Observation 3

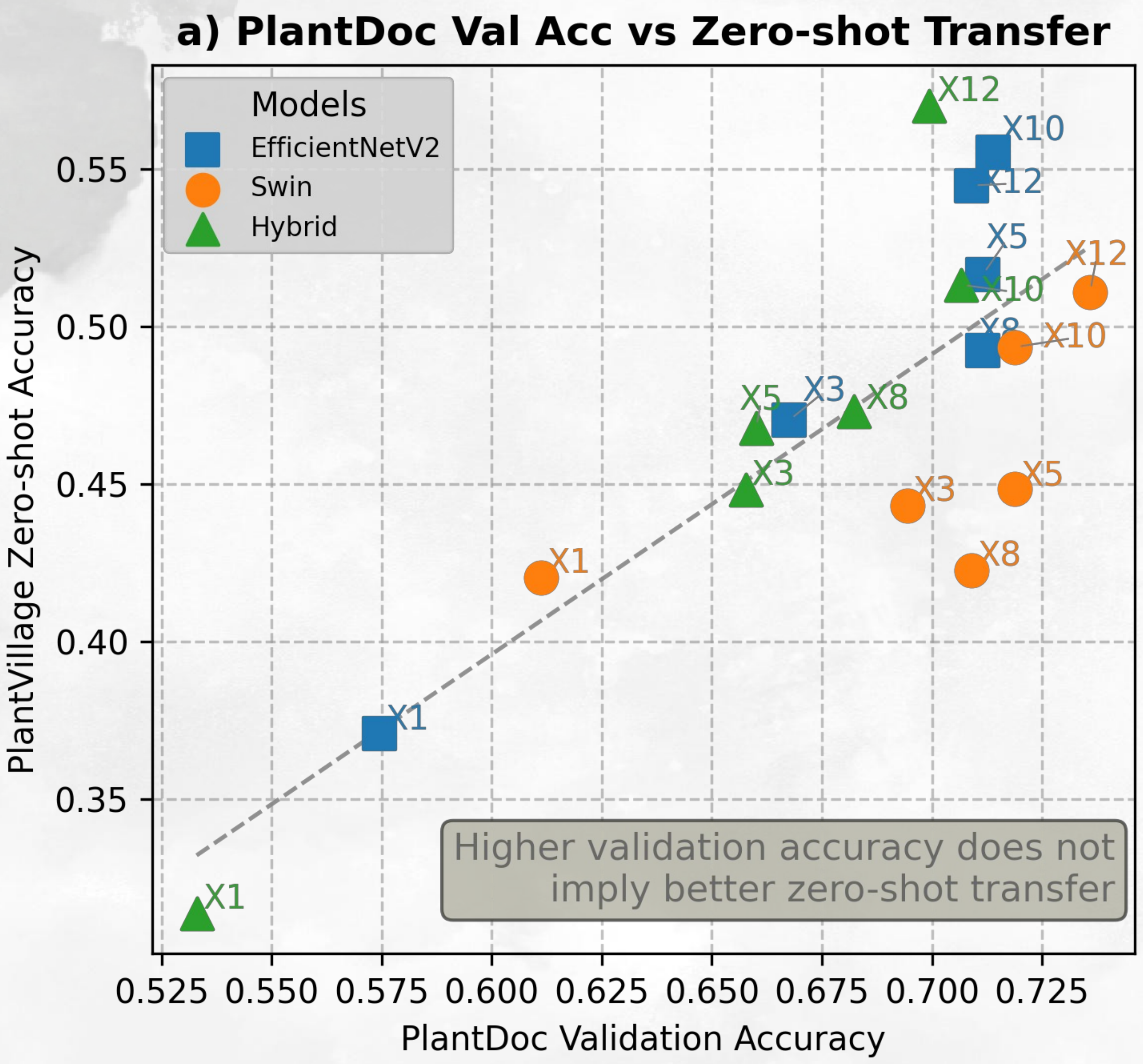
- EfficientNetV2 and Swin **declines** (−2.40% and −2.58%),
- Hybrid model **improves** (+0.54%)

Observation 4

- Hybrid model performance **scale effectively with increasing augmentation**,
- Starts with **lowest accuracy** at x1 (0.31)
- **Surpasses all** other architectures at the x12 (0.57)

EXPERIMENTAL RESULTS

Validation Paradox



Experiment Observations:

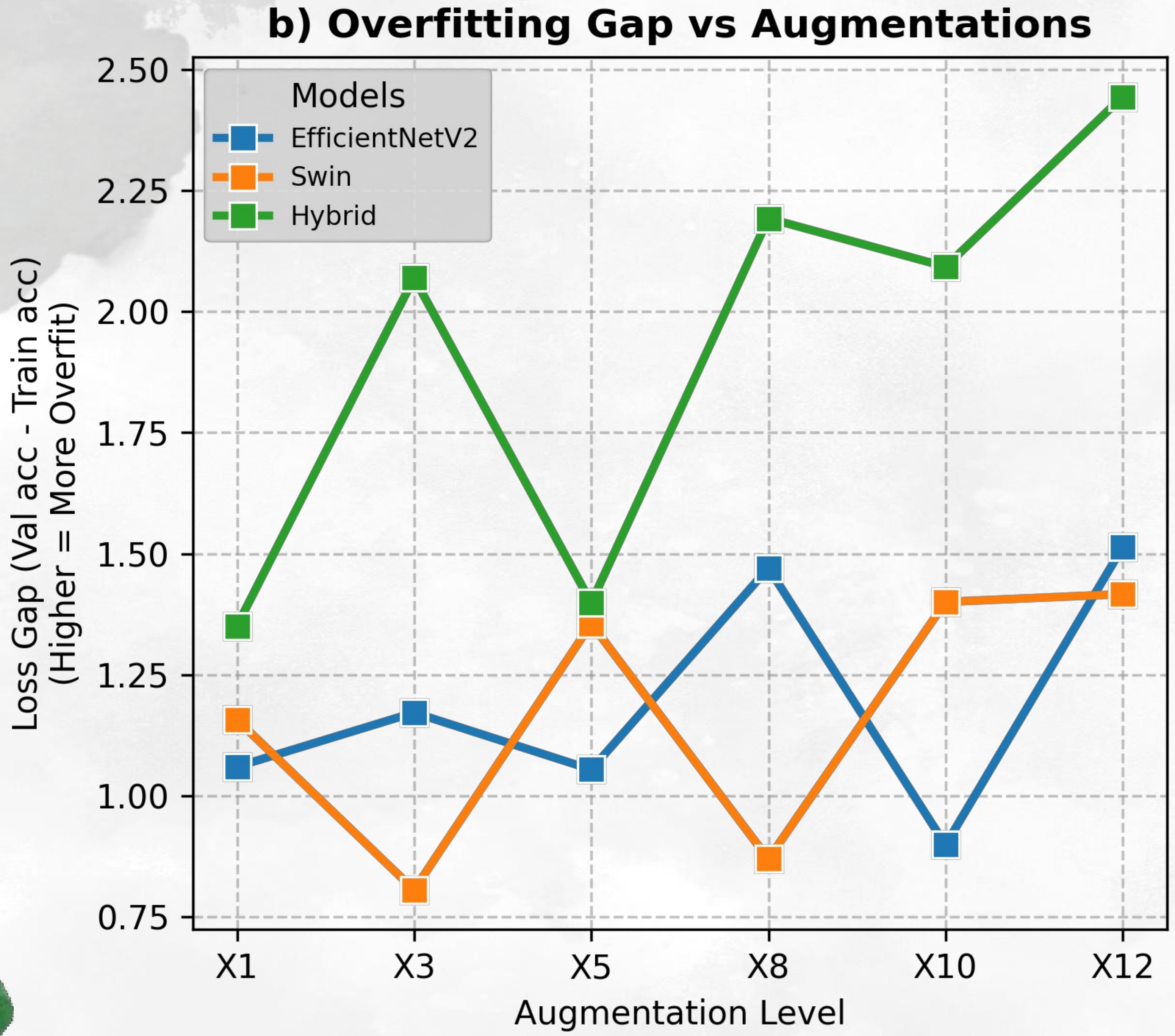
- At x12, Swin achieved **peak validation** (0.73), but lower zero-shot transfer (0.51)
- Hybrid model achieved the **peak zero-shot** (0.58) despite having significantly **lower validation** score (0.70)
- EfficientNetV2 maintains a balanced performance cluster near higher zero-shot metrics at x5, x10, and x12

Paradox:

- **Higher validation** should lead to **better generalization**

EXPERIMENTAL RESULTS

Analyzing the Domain Gap & Overfitting



Observations:

- **Hybrid model** consistently displays the **largest loss gap**, peaking at 2.48 at the x12 augmentation level

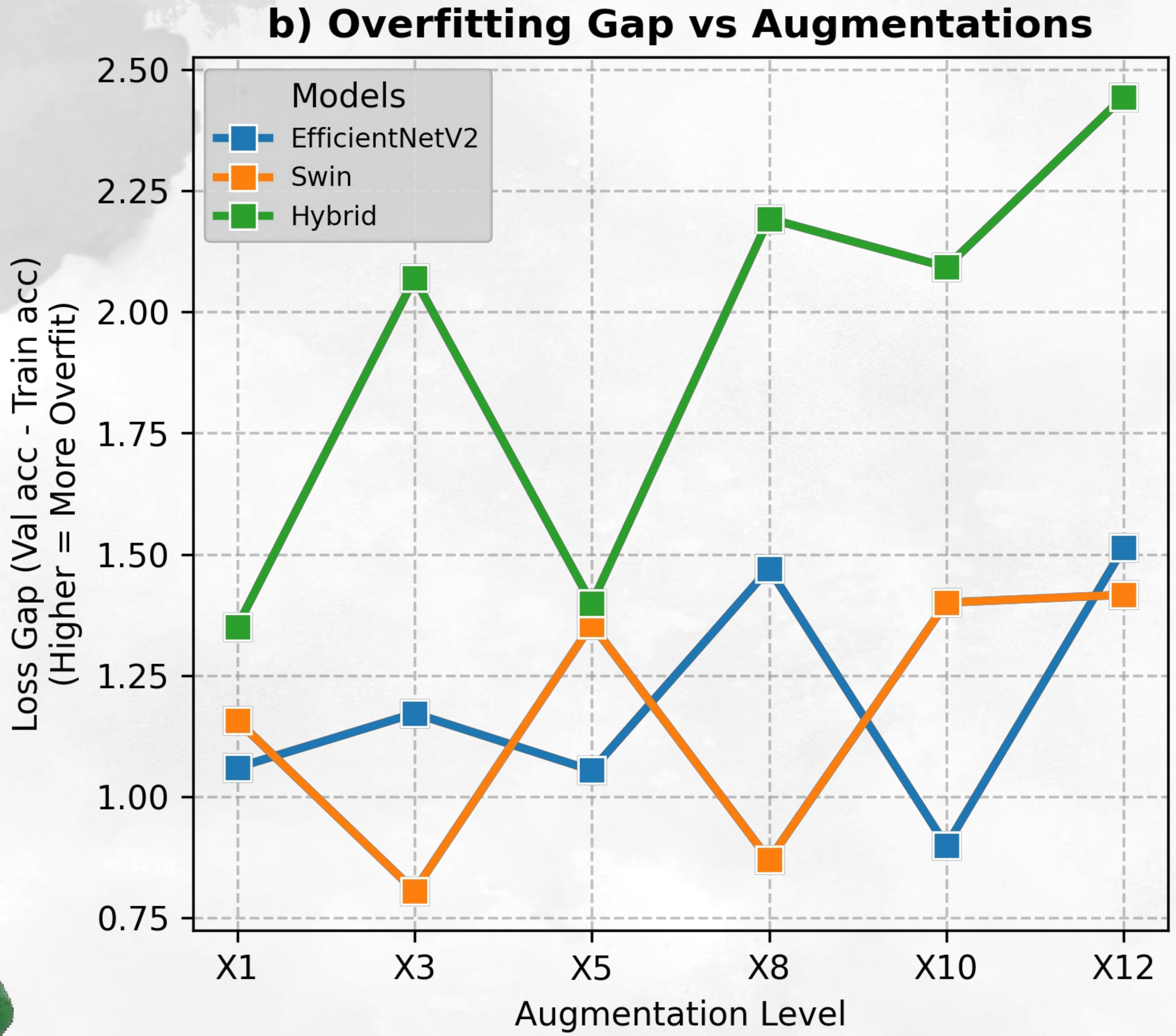
Paradox:

- By conventional logic, this **larger gap** should lead to **poor performance**

- However, Hybrid model achieved highest zero-shot results at x12

EXPERIMENTAL RESULTS

Analyzing the Domain Gap & Overfitting



Why this happens?

Advantages of architecture

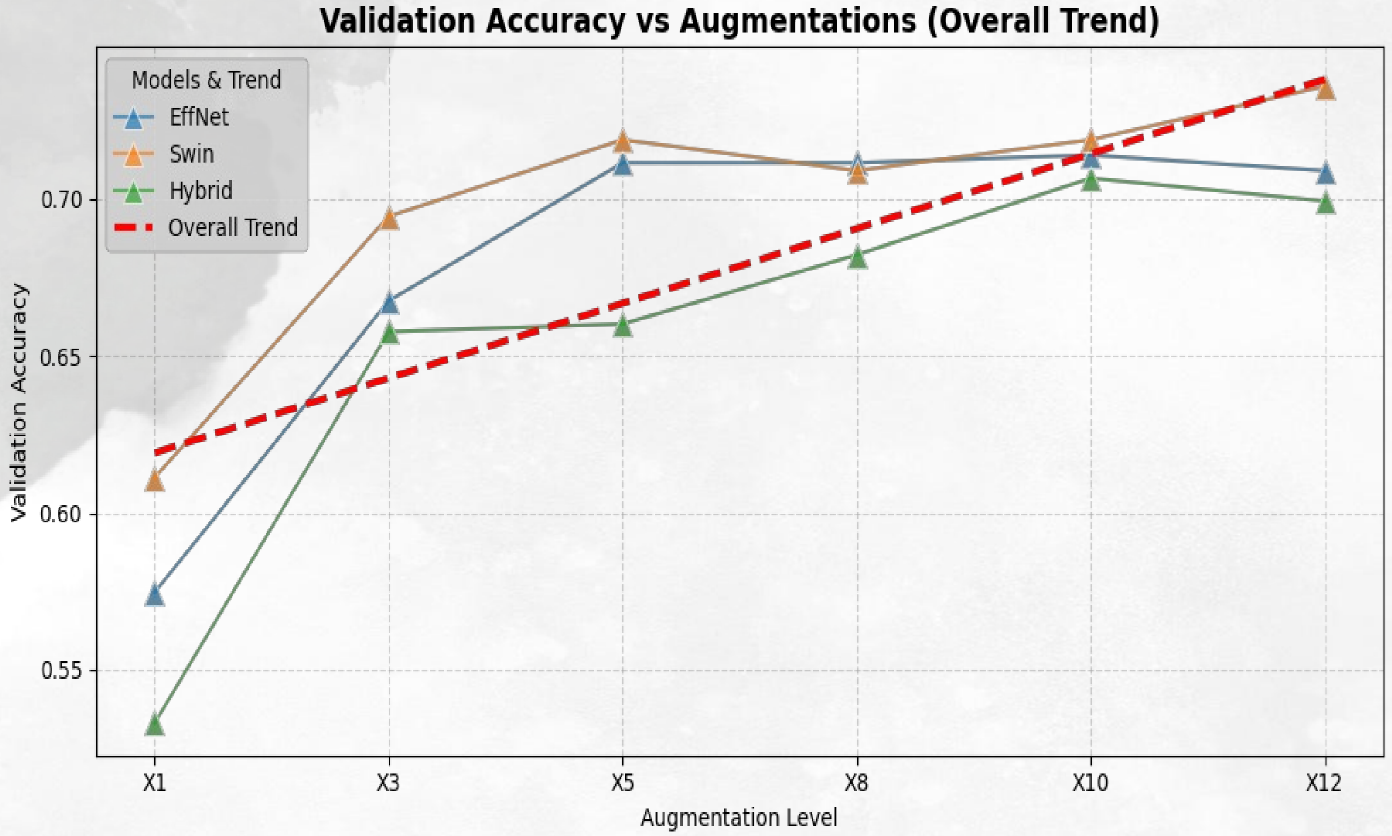
- CNN act to capture local texture pathology
- Transformer global attention act as a filter for environmental noise

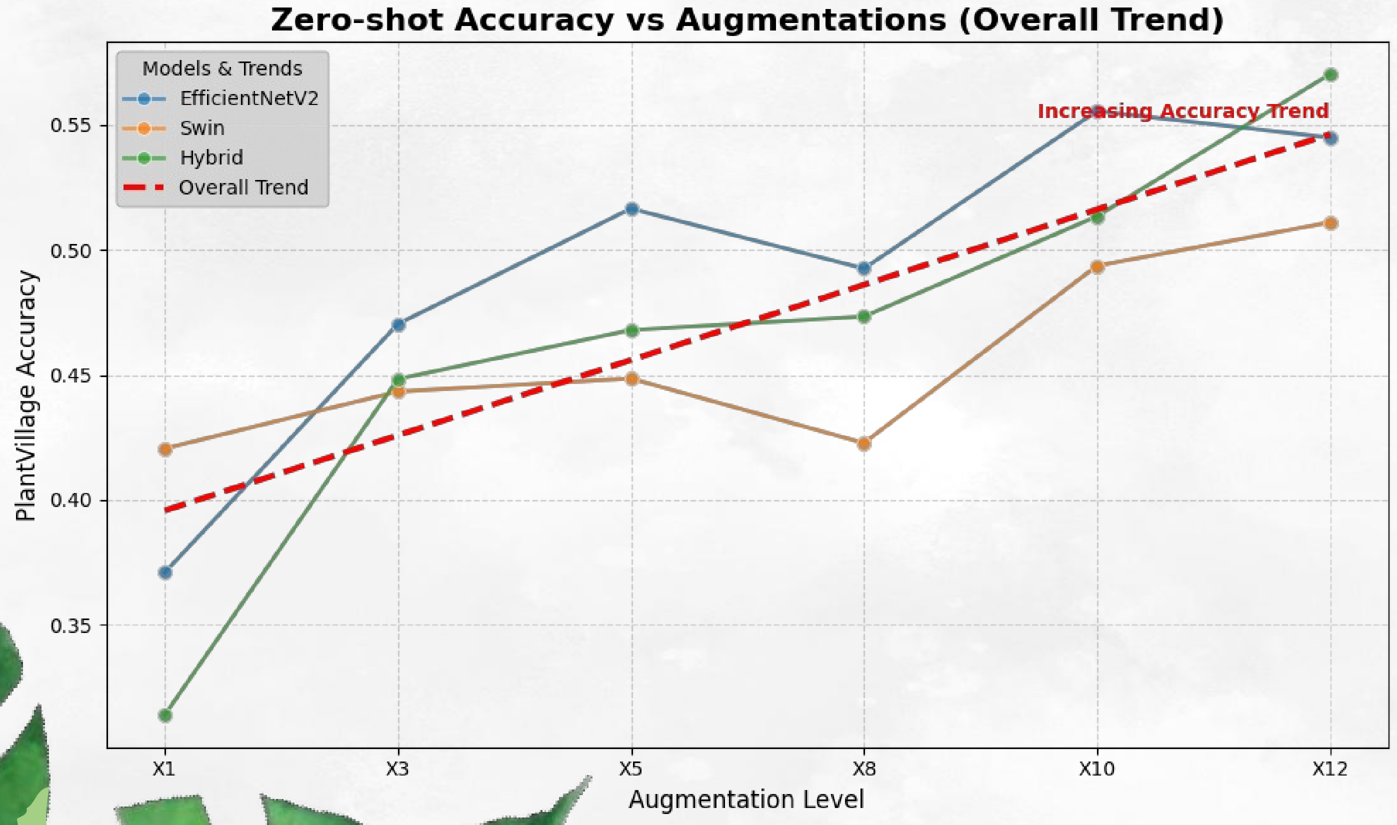
Advantages of aggressive augmentation

- Higher parameter model was forced to ignore noises correlations
- Isolate to focus on morphology (feature) of the disease
- Act as a regularization that allow complex architecture to bridge the gap between dirty and clean domain

EXPERIMENTAL RESULTS

Performance Summary





- Scaling of augmentation show an upward trend in validation and zero-shot accuracy.

- **For noisy real-world classification task,**
  - Swin perform the best across all augmentation, but at x8, Swin performance drops, causing EfficientNetV2 to surpass it
  - Hybrid perform the worst across all augmentation

- **For noisy to clean transfer,**
  - Swin perform the best with x1 augmented dataset, but the worst for higher augmentation
  - EfficientNetV2 perform the best at moderate augmentation (x3 to x10)
  - Hybrid perform better with high intensity augmentation (x12)

CONCLUSION

FINDINGS

- Augmentation scaling plays a critical role in generalisation
- EfficientNetV2: performs better with lower augmentation
- Swin: struggles with noisy data due to its global attention
- Hybrid: requires larger augmentation, but achieves best performance

FUTURE WORKS

- Investigate convergence point across more varied models
- Explore other adaptation technique

THANK YOU